Intro to Machine Learning
Practice Problems 4: QDA and Naive Bayes

Practice problems, not turned in

Directions: These practice problems go with Week 3 (quadratic discriminant analysis and naive Bayes). They are not turned in, but the quizzes draw on them, so work through each one before looking at the solutions. Problems adapted from ISL cite the exercise; see the book for the original wording. Practice Problems 3 covers logistic regression and LDA.

Notation follows ISL 4.4. Classes are $k = 1, \dots, K$, $\pi_k = \Pr(Y = k)$ is the prior, $f_k(x)$ is the density of X in class k, and $p_k(x) = \Pr(Y = k \mid X = x)$ is the posterior. QDA assumes $X \mid Y = k \sim N(\mu_k, \Sigma_k)$; naive Bayes assumes $f_k(x) = \prod_{j=1}^{p} f_{kj}(x_j)$.

1. **QDA is quadratic.** (ISL 4.8, Exercise 3; ISL 4.4.3, ESL 4.3.) Take $p = 1$ and $X \mid Y = k \sim N(\mu_k, \sigma_k^2)$, with a different variance in each class. Find the discriminant $\delta_k(x)$ the Bayes classifier maximizes, and show that the decision boundary between two classes is not linear in x unless $\sigma_1 = \sigma_2$.

2. **QDA by hand: same means, different spreads.** (ISL 4.4.3.) Two equally likely classes have $X \mid Y = 1 \sim N(0, 1)$ and $X \mid Y = 2 \sim N(0, 4)$.

   (a) Find the Bayes decision rule. Where is each class predicted?

   (b) Now let $\pi_1 = 0.8$. How does the rule change?

3. **LDA or QDA?** (ISL 4.8, Exercise 5; ISL 4.4.3, 4.5.) Answer each and explain.

   (a) If the Bayes decision boundary is linear, do we expect LDA or QDA to do better on the training set? On the test set?

   (b) If the Bayes decision boundary is non-linear, do we expect LDA or QDA to do better on the training set? On the test set?

   (c) In general, as n increases, do we expect the test accuracy of QDA relative to LDA to improve, decline or stay the same?

   (d) True or false: even if the Bayes boundary is linear, QDA will probably achieve a lower test error rate than LDA, because QDA is flexible enough to model a linear boundary.

   (e) With $p = 10$ predictors and $K = 3$ classes, how many covariance parameters do LDA, QDA and Gaussian naive Bayes each estimate? How many observations does each class need for QDA's $\hat{\Sigma}_k$ to be invertible?

4. **Naive Bayes with categorical predictors, by hand.** (ISL 4.4.4.) Twelve students took a quiz. For each we record whether they attended the lecture, how many of the practice problems they did (none, some or all), and whether they slept at least 7 hours:

Student	Result	Attended	Practice problems	Slept 7h+
1	pass	yes	all	yes
2	pass	yes	all	no
3	pass	yes	some	yes
4	pass	yes	some	yes
5	pass	no	all	yes
6	pass	yes	none	yes
7	pass	no	some	no
8	pass	yes	some	no
9	fail	no	none	no
10	fail	yes	none	no
11	fail	no	some	yes
12	fail	no	none	no

(a) Estimate the priors, and for each class the proportion of students in each category of each predictor.

(b) Student A attended, did some of the practice problems and slept less than 7 hours. Find the naive Bayes estimate of $\Pr(\text{pass} \mid x)$. (The fractions work out exactly.)

(c) Student B did not attend, did none of the problems and slept less than 7 hours. Find $\Pr(\text{pass} \mid x)$.

(d) Student C did not attend, did all of the problems and slept less than 7 hours. What does naive Bayes give, and why is it a problem? Is C's combination of predictor values in the training data?

(e) Redo (d) with Laplace smoothing: add 1 to every count, so a predictor with m categories has $\hat{f}_{kj}(v) = (\text{count} + 1)/(n_k + m)$. Leave the priors unsmoothed.

(f) Without the naive Bayes assumption, how many probabilities would you have to estimate per class to describe the joint distribution of the three predictors? How many does naive Bayes estimate? Why does that matter with 12 students?

5. **What naive Bayes assumes, and when it works.** (ISL 4.4.4, 4.5.) Answer each in a few sentences.

(a) State the naive Bayes assumption in words. Is it about how the predictors relate to each other in the population as a whole, or within each class? Give an example of two predictors that are strongly correlated in the population but roughly independent within each class.

(b) The naive Bayes assumption is rarely true, yet naive Bayes often classifies well. Explain why, in terms of bias and variance. In what kind of problem does it tend to beat QDA?

(c) A spam filter uses the words "free" and "offer", which almost always appear together in spam. How does naive Bayes treat the two words, and what does that do to its estimate of $\Pr(\text{spam} \mid x)$ for an email containing both? Does it necessarily change which class is predicted?

(d) True or false: Gaussian naive Bayes, with a separate variance for each predictor in each class, has a linear decision boundary.